

Diet and Diabetes: Recipes for Success

PhysiciansCommittee
for Responsible Medicine



Diabetes Basics

In the past few years, much of what we thought we knew about diabetes has been turned on its head. New understanding of the nutritional causes of diabetes gives us the power to keep it from occurring or to turn it around.

Here is what is supposed to happen: Our bodies turn starchy and sugary foods into glucose for our muscle cells to use for fuel. Insulin, a hormone made in the pancreas, ushers glucose into the cells. People with type 2 diabetes, the most common type, generally have enough insulin. However, their cells become resistant to it, leaving too much glucose in the bloodstream, where it can cause problems.

Over the short run, people with uncontrolled diabetes may feel tired and thirsty, urinate frequently, and notice blurred vision. In the long run, they are at risk for heart disease, kidney problems, vision loss, nerve damage, and other difficulties.

Dietary Approaches to Diabetes

Diabetes diets typically call for portion control, carbohydrate counting, and, for those who are overweight, calorie restrictions. Fortunately, there is another way. **Low-fat, plant-based diets are ideal for diabetes** and the conditions associated with it, such as heart disease, weight gain, high cholesterol, and high blood pressure. And they offer the advantage of not requiring any weighing or measuring of portions. Going hungry is not necessary!

The old approach recommended cutting down on carbohydrates. It's true that overly processed carbohydrates—those made with sugar or white flour, for example—are poor choices. However, unprocessed or minimally processed foods, such as potatoes, rice, oats, beans, pasta, fruit, and vegetables, were the main part of the diet in countries where people were traditionally fit and trim and where diabetes was rare. Unfortunately, highly processed carbohydrates and affordable meat and cheese dishes have moved in, and now we have a worldwide type 2 diabetes epidemic.

Because our bodies are built to function best on high-fiber carbohydrates, a low-fat plant-based approach to diabetes management is the most impactful. Fat from animal products, fried foods, and oils interferes with insulin's ability to move glucose into the cells.¹ Eating less fat not only reduces body fat but also allows insulin to work better, reducing blood sugar and addressing the root cause of diabetes. **However, choosing skinless chicken, skim milk, and baked fish is not enough of a change for most people to beat diabetes.**

The new approach eliminates fatty foods and animal protein, such as meats, dairy products, and oils, and offers unlimited grains, legumes, fruits, and vegetables. In 2023, researchers showed that a whole food, plant-based diet can improve insulin function in as little as seven days. Individuals with type 2 diabetes, dependent on injecting insulin, followed either a Dietary Approaches to Stop Hypertension (DASH) diet or a plant-based diet. The plant-based diet was more effective at lowering injected insulin doses, improving insulin function and blood sugars, all within just seven days.² Research shows that patients can also significantly reduce their oral medication needs and maintain their gains even years later.^{3,4} The dietary changes are simple, but profound, and they work.

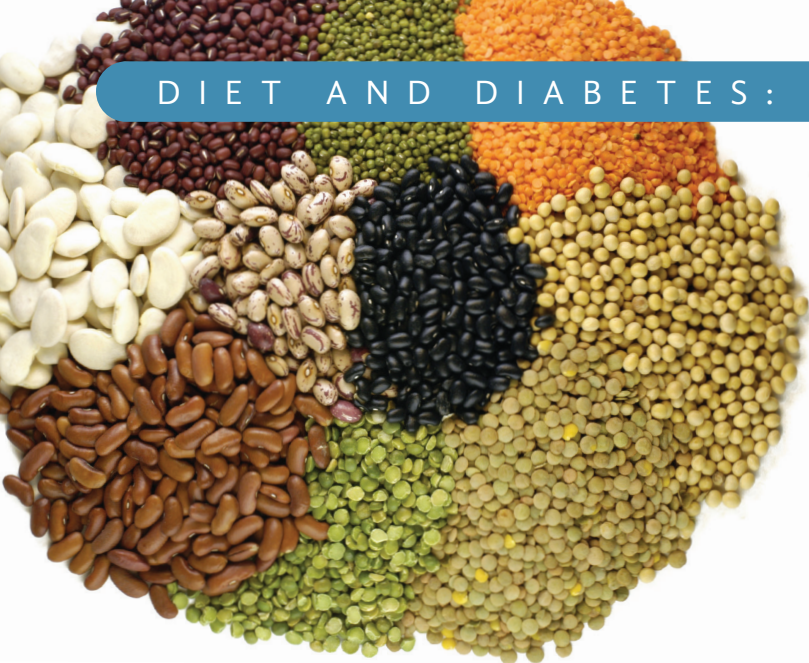
As technology improves, scientists are able to look inside the muscle and liver with MRIs. What they've found is that people

Success Story

Former University of Michigan Football Player Tackles Diabetes



In 2009, Marc's doctor told him he would be on insulin for the rest of his life. His entire family struggled with chronic diseases, which had led to blindness and the need for amputations, organ transplants, and dialysis, so Marc believed that this was his fate, too. He was sick for over a decade and took five medications daily, including insulin injections. He tried eating less, exercising more, and counting carbohydrates, but nothing seemed to work. One day he read *Dr. Neal Barnard's Program for Reversing Diabetes* and began a low-fat, plant-based diet. In just two months he was able to eliminate all his medications, and in three months he had lost 50 pounds. Today he thrives and lives an amazing life free of medication and disease. If Marc can do it, so can you!



who change to a low-fat, vegan diet can reduce fat in their liver and muscle cells by 34% and 10%, respectively.⁵ Researchers have also found that the cells that secrete insulin can start working better after following a low-fat plant-based diet.⁶ Together, these studies show that it's possible to reverse the underlying cause of type 2 diabetes.

A study conducted by the Physicians Committee for Responsible Medicine with the George Washington University and the University of Toronto looked at the health benefits of a low-fat, vegan diet (excluding all animal products) in people with type 2 diabetes.⁷ Portions of vegetables, grains, fruits, and legumes were unlimited. The vegan diet group was compared with a group following a portion-controlled, higher-fat diet based on American Diabetes Association (ADA) guidelines. The results of this 22-week study were astounding:

- A total of 43% of the vegan group and 26% of the ADA group reduced their diabetes medications. Among those whose medications remained constant, the vegan group lowered hemoglobin A1C, an index of long-term blood glucose control, by 1.2 points, three times the change in the ADA group.
- The vegan group lost an average of about 13 pounds, compared with about 9 pounds in the ADA group.
- Among those participants who didn't change their lipid-lowering medications, the vegan group also had more substantial decreases in their total and LDL cholesterol levels compared with the ADA group.

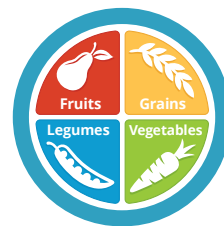
This study illustrates that a plant-based diet can dramatically improve the health of people with diabetes. It also shows that people found this way of eating highly acceptable and easy to follow. Both the American and Canadian Diabetes Associations include plant-based as a diet option, and the American Association of Clinical Endocrinologists and the American College of Endocrinology recommend a plant-based meal plan for people with type 2 diabetes.

The Remarkably Simple New Dietary Approach to Diabetes

Give It a Three-Week Trial. What Have You Got to Lose?

1. Build Your Meals From the Power Plate.

It's not complicated! Fill your plate with whole grains, legumes (beans, lentils, peas), fruits, and vegetables. No need to avoid fruit. In fact, people who eat the most fruit are at a lower risk for diabetes.⁸ Drink water. Keep nuts or seeds to a small handful once a day.



2. Begin a Vegan Diet: Avoid Animal Products.

A vegan diet has no animal products at all: no red meat, poultry, pork, fish, dairy products, and eggs. Why? Animal products contain saturated fat, which is linked to heart disease, insulin resistance, and certain forms of cancer. They also contain cholesterol and, of course, animal protein. It may surprise you to learn that diets high in animal protein can aggravate kidney problems. All the protein you need can be found in whole grains, legumes, and vegetables.

3. Avoid Added Vegetable Oils and Other High-Fat Foods.

Although vegetable oils are healthier than animal fats, oils are not health foods. All fats and oils are high in calories; 1 gram of any fat or oil has 9 calories, while 1 gram of carbohydrate has only 4 calories. The amount of fat we really need each day is quite small and comes packed inside the Power Plate vegetables, grains, and beans.

Avoid oily sauces and salad dressings and foods fried in oil. Limit olives, avocados, nuts, and peanut butter. Read labels, and choose mostly foods with no more than 2-3 grams of fat per serving.

4. Favor Foods With a Low Glycemic Index.

The glycemic index (GI) identifies foods that raise blood sugar more than other foods. High-GI foods can also raise triglyceride levels. Fortunately, beans, oats, sweet potatoes, and, surprisingly, white and wheat pasta are among foods that are lower-GI champions. So are breads such as pumpernickel, rye, multigrain, sourdough, and tortillas. Lower-GI cereals are bran cereals, muesli, and rolled or steel-cut oats. Grains such as barley, brown rice, couscous, corn, and quinoa have a low GI. High-GI foods to limit are sugar and sugary products, white and wheat bread, cornflakes, and puffed rice cereals.



5. Go High-Fiber.

Aim for at least 40 grams of fiber each day. Choose beans, vegetables, fruits, and whole grains (e.g., whole-wheat pasta, barley, oats, quinoa). Aim for at least 3 grams of fiber per serving on labels and 10 to 15 grams per meal. Start slowly. Expect a change in bowel habits (usually for the better). Gassiness from beans can be minimized with small servings and thorough cooking and, if a problem, will get better over time!

A note on vitamin B12: Those following a diet free of animal products (and all adults over the age of 50) should take a B12 supplement to protect blood and nerve cells.

Additional Resources

- PCRM.org/Diabetes: Learn more about eating to manage or reverse diabetes.
- NationalDiabetesProgram.org: Join a free diabetes support group.
- 21DayKickstart.org: Free mobile app with meal plans, recipes, and a monthly tracking tool.
- PCRM.org/FFL: Take a Food for Life nutrition and cooking class.
- Watch an inspiring documentary: *Forks Over Knives*, *The Game Changers*, *What the Health?*, and *Eating You Alive*.
- Helpful books and cookbooks:
 - *Dr. Neal Barnard's Program for Reversing Diabetes*
 - *Dr. Neal Barnard's Cookbook for Reversing Diabetes*
 - *Kick Diabetes Essentials* by Brenda Davis, RD

Pasta With Lentil Marinara Sauce

Makes 5 servings

- 1 pound pasta of choice
- 1 jar (26 ounces) fat-free, low-sodium, tomato-based pasta sauce
- 1 can (15 ounces) lentils, rinsed and drained
- 1/2 cup dry red wine (can be nonalcoholic) or low-sodium vegetarian broth
- Salt, to taste
- Freshly ground black pepper

Cook the pasta according to package directions.

Meanwhile, combine the pasta sauce, lentils, and wine or broth in a medium saucepan. Heat gently and season with the salt and pepper. Serve over the drained pasta.

Per serving: 470 calories, 19 g protein, 91 g carbohydrate, 9 g sugar, 2 g total fat, 3% calories from fat, 0 mg cholesterol, 8 g fiber, 173 mg sodium

Black Bean and Corn Salad With Lime

Makes 4 servings

This delicious salad is easy to make and will keep you full for hours. It's a great recipe share with friends at a potluck.

- 1 15-ounce can (840 g) black beans, rinsed and drained
- 1 small yellow bell pepper, chopped
- 1 small red bell pepper, chopped
- 1 small tomato, chopped
- 1 cup (166 g) frozen corn, thawed
- 2 scallions, chopped
- 1 tablespoon (15 mL) fresh cilantro
- 1 teaspoon dried oregano
- 1 teaspoon dried basil
- 1/2 teaspoon ground cumin
- Juice of 1/2 lime
- 1/4 teaspoon iodized salt
- 1/8 teaspoon black pepper

Combine all ingredients in a salad bowl. Add additional lime juice, cilantro, salt, or pepper to taste. Serve chilled or at room temperature.

Per serving (1/4 of recipe): 154 calories, 8 g protein, 31 g carbohydrate, 4 g sugar, 1 g total fat, 5% calories from fat, 0 mg cholesterol, 10 g fiber, 291 mg sodium

Source: *The Get Healthy, Go Vegan Cookbook* by Neal Barnard, MD; recipe by Robyn Webb

Berry Mousse

Makes 4 servings

This is so easy that it's hardly a recipe! Your blender does most of the work. This can be eaten as a pudding or used as a topping for fruit.

- 1 package (12.3 ounces) reduced-fat, extra-firm silken tofu, crumbled
- 2 3/4 cups thawed frozen unsweetened berries of choice
- 3 tablespoons sugar or 2 tablespoons agave nectar
- 1 tablespoon berry liqueur (optional)

Blend the tofu, berries, sugar or agave nectar, and liqueur, if using, in a blender or food processor until smooth. Spoon into 4 pudding dishes and refrigerate until chilled.

Per serving: 123 calories, 7 g protein, 24 g carbohydrate, 17 g sugar, 1 g total fat, 5% calories from fat, 0 mg cholesterol, 3 g fiber, 89 mg sodium

Recipes from *Dr. Neal Barnard's Program for Reversing Diabetes: The Scientifically Proven System for Reversing Diabetes Without Drugs* by Neal D. Barnard, MD. Recipes by Bryanna Clark Grogan. ◀

Meal Suggestions

Breakfast

- Hot cereals: oatmeal with cinnamon, berries, or applesauce
- All-Bran or muesli with nondairy milk and fresh fruit
- Fresh fruit or a smoothie
- Pumpernickel or rye toast topped with jam
- Sweet potato hash with sauteed mushrooms, peppers, and onions
- Tofu scramble

Lunch

- Mixed-vegetable salad with lemon juice, low-fat dressing, or soy or teriyaki sauce
- Legume-based salads: three-bean, chickpea, lentil, or black bean and corn
- Grain-based salads: noodle, couscous, bulgur, or rice
- Soups: carrot ginger, mixed vegetable, black bean, vegan chili, lentil, minestrone, split pea, etc.
- Hummus spread on whole-wheat pita with grated carrots, sprouts, and cucumbers
- Black bean and sweet potato burrito with corn and tomatoes
- Sandwich made with low-fat meat alternatives such as barbecue seitan, Lightlife or Yves products, and your favorite sandwich veggies

Dinner

- Pasta marinara: can be made with many commercial sauces (any brand that has less than 3 grams of fat per serving and is free of animal products)
- Beans and rice: black beans with salsa, vegetarian baked beans, or low-fat refried beans

- Soft tacos: a flour tortilla filled with beans, lettuce, tomato, and salsa
- Fajitas: lightly sauteed sliced bell peppers, onion, and eggplant with fajita seasonings
- Chili: homemade or vegan boxed or canned versions
- Veggie lasagna: seasoned tofu to replace the ricotta cheese, layered with grilled veggies
- Vegetable stir-fry: vegetables seasoned with soy sauce or other low-fat stir-fry sauce and served over pasta, beans, or rice

Snacks

- Fruit
- Carrot, celery, or other vegetables with low-fat hummus
- Baked tortilla chips with salsa or bean dip
- Air-popped popcorn or rice cakes
- Toast with jam
- Pretzels



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This fact sheet is not intended as a comprehensive program for diabetes. Be aware that a change in diet can produce big results. For some, there is a risk that low blood sugar can occur if diabetes medications are not lowered or eliminated. If you have diabetes, consult your health care provider and tailor a program for your needs.